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THINGBD · PROJECT DOCUMENTATION

ThingBD CRM

The complete workflow guide

How the automation CRM works from the first lead to the closed deal: every module, who does what, what runs in the background, and how it is deployed, backed up and operated.

PHASES	STACK	TESTS	VERSION
0-8 complete	Django 5.2 + Celery	301 passing	29 Oct 2026

github.com/Rabbi-2529/automation_crm · thingbd.com

CONTENTS

What's in this guide

#	Section	Covers
1	The product at a glance	what the CRM does, the modules, the menu
2	Roles and permissions	agent, manager, admin
3	System architecture	how the pieces fit together
4	The core workflow: lead to revenue	the end-to-end journey
5	The event engine	why everything reacts to everything
6	Module workflows	step by step, every screen
7	Automation reference	triggers, conditions and actions
8	A day in the life	routines per role
9	First-time setup	the admin checklist
10	Background jobs	what runs on its own, and when
11	Deployment and operations	environments, deploy, backups, security
12	Demo access and project history	logins, phases, commits

SECTION 1

The product at a glance

ThingBD CRM is a multi-workspace CRM with a built-in automation engine. A business (a **workspace**) keeps its contacts, companies and deals in one place, talks to customers by email, SMS and WhatsApp from a shared inbox, and lets workflows do the repetitive work: assigning leads, sending follow-ups, creating tasks and moving deals. Every workspace's data is kept strictly separate.

Area	What it gives the team
Contacts & companies	Profiles, tags, custom fields, notes, files, full timeline, duplicate warnings, CSV import/export, bulk edits.
Deals & pipeline	Kanban board per pipeline, drag between stages, won/lost with reasons, weighted forecast, rotting deals.
Tasks & calendar	Calls, emails, meetings and follow-ups with due times and email reminders; drag-to-reschedule calendar; Google Calendar feed.
Inbox	Email, SMS and WhatsApp conversations in one place; templates with variables; open and click tracking; AI reply drafts.
Automations	Trigger, conditions, actions; waits; round-robin assignment; run logs for every step; incoming and outgoing webhooks.
Lead management	Embeddable web forms, Facebook lead ads, lead scoring with hot/warm/cold grades, speed-to-lead alerts.
Marketing	Segments, multi-step email/SMS sequences that stop on reply, broadcasts, unsubscribe and STOP handling, quiet hours.
Reports	Revenue, win rate, funnel, forecast, lost reasons, lead sources, rep leaderboard, targets, scheduled report emails, CSV.

Area	What it gives the team
Integrations	REST API with keys and docs, signed webhooks, WhatsApp Cloud API, Facebook lead ads, iCalendar feed.
Help & AI	Help center, in-app AI assistant (free open-source models by default) and AI drafted replies, always reviewed by staff.

The sidebar

Dashboard and **Help & AI assistant** sit at the top, followed by three groups that open on click: **CRM** (Contacts, Companies, Imports, Deals, Tasks, Calendar, Inbox, Automations, Reports), **Marketing** (Segments, Sequences, Broadcasts, Web forms, Lead scoring) and **Workspace** (Teams, Members, Tags, Message templates, Pipelines, Settings, AI assistant, Integrations, Messaging channels, Custom fields, System status). The group holding the current page opens by itself; a red dot on a closed group means overdue tasks or unread messages.

SECTION 2

Roles and permissions

Each person has one role per workspace. Higher roles include everything below them. A person can belong to several workspaces and switch between them from the top bar.

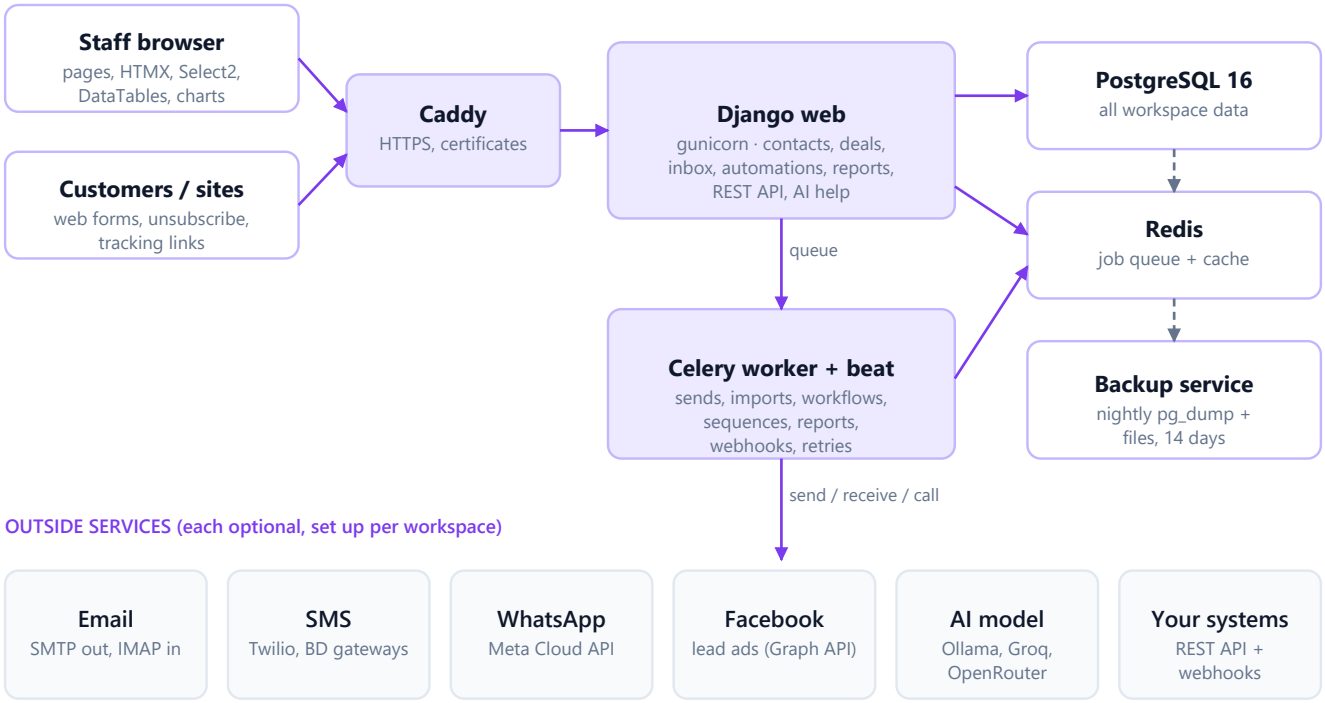
What	Agent	Manager	Admin
Work with contacts, companies, deals, notes, files, tasks; move, win and lose deals; bulk tag / owner / stage	Yes	Yes	Yes
Send email / SMS / WhatsApp, use the inbox, log calls, ask the AI assistant, see reports	Yes	Yes	Yes
Delete records, bulk delete, CSV import and export		Yes	Yes
Tags, pipelines, lost reasons, message templates, web forms, lead scoring, targets, report exports		Yes	Yes
Members and roles, custom fields, workspace settings			Yes
Messaging channels, integrations (API keys, webhooks, Facebook), AI assistant settings, system status			Yes

> Notes and files can be deleted by their author or any manager. A task can be edited by its assignee, its creator or a manager; anyone can tick it off. API keys act with the role of the person who created them.

SECTION 3

System architecture

The app is one Django project. Web requests are answered by the web service; anything slow or scheduled (sending messages, imports, workflow steps, retries, report emails) goes through Redis to the Celery worker, and Celery beat starts the scheduled jobs.

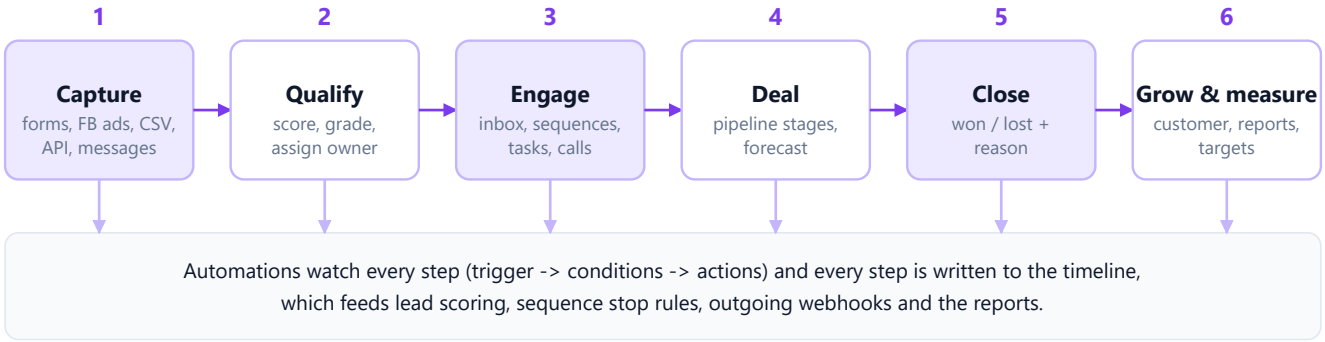


Layer	Technology	Notes
Back end	Django 5.2, Python 3.12	15 apps under apps/: core, accounts, orgs, contacts, activities, pipeline, tasks, messaging, automation, leads, campaigns, reports, integrations, assistant.
Data	PostgreSQL 16 (SQLite for quick local runs)	Every tenant record carries its workspace and is always queried through it.
Jobs	Celery + Redis	Locally tasks can run inline (eager mode) so Redis isn't needed.
Front end	Django templates, Tailwind, HTMX	Select2 on every select, DataTables (server-side for big lists), FullCalendar, Chart.js.
API	Django REST Framework + OpenAPI	Interactive docs at /api/docs/.
Production	Docker Compose, gunicorn, Caddy	Automatic HTTPS, compiled CSS and vendored JS, nightly backups.

SECTION 4

The core workflow: lead to revenue

Every feature in the CRM serves one journey: turning a new enquiry into a paying, measured customer. This is that journey end to end.



Step 1 · Capture the lead

- **Web form** on your website (Marketing → Web forms): a submission creates or updates the contact; existing data is never overwritten, only blanks filled.
- **Facebook / Instagram lead ad**: the lead arrives through Meta's webhook and becomes a contact with source Facebook lead ad.
- **Incoming message**: an email, SMS or WhatsApp from a known number or address is attached to that contact; unknown senders can be saved as contacts.
- **CSV import** (CRM → Imports), the **REST API** or a **workflow webhook** from another system, or a person adding it by hand.

Step 2 · Qualify and assign

- Lead scoring adds points for who the person is (job title, source, custom fields) and what they do (opens, clicks, replies, form posts). Scores become Hot, Warm or Cold.
- A workflow assigns the owner (a specific person or round robin across a team) and can tag, set the lifecycle stage and create the first task.
- **Speed to lead**: if nobody emails, messages or calls a new lead within N hours, a workflow can alert the manager or reassign.

Step 3 · Engage

- Reps reply from the contact page or the inbox, log calls and plan tasks; the AI assistant can draft a reply for them to edit.
- Sequences send a planned series of emails/SMS and stop by themselves when the person replies; broadcasts reach a whole segment.
- Quiet hours hold automated messages until the sending window opens; unsubscribed contacts are always skipped.

Step 4 · Work the deal

- Create a deal from the contact or the board (CRM → Deals). The contact becomes an Opportunity automatically.
- Drag the card through the stages; each stage has a win probability that drives the weighted forecast. Deals idle too long are flagged.
- Workflows react to stage changes (send a proposal email, create a task for the manager, notify the team).

Step 5 · Close

- Mark the deal **Won** (the contact becomes a Customer) or **Lost** with a reason. Closed deals can be reopened.
- Won and lost events can start workflows and are pushed to other systems (accounting, invoicing) through outgoing webhooks.

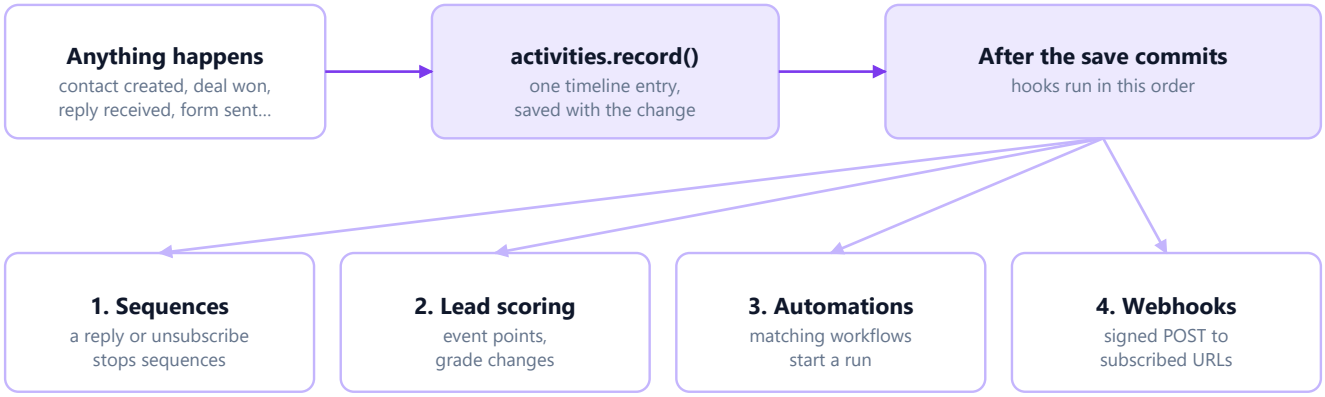
Step 6 · Grow and measure

- **CRM → Reports** shows revenue, win rate, funnel, forecast, lost reasons and lead sources; the leaderboard compares reps against their targets.
- Managers and reps receive scheduled report emails (team summary or personal pipeline digest).
- Date-based workflows (renewals, birthdays) and the contact inactive trigger bring customers back.

SECTION 5

The event engine

The reason the CRM feels connected: every change is written to the timeline through one function, `activities.services.record()`. When the database save is committed, four hooks react to the new timeline entry.



Timeline events (examples)	Who listens
Created, updated, lifecycle stage changed, owner changed, tag added / removed, note, file, imported	Automations, webhooks
Deal moved, won, lost, reopened · task created / completed	Automations, scoring, webhooks, reports
Email sent / received / opened / clicked · SMS and WhatsApp sent / received · call logged · message failed	Sequences (stop on reply), scoring, automations, webhooks
Form submitted · Facebook lead ad · lead grade changed	Automations, scoring, webhooks
Unsubscribed / resubscribed · added to / left a sequence	Sequences, scoring, webhooks

> Because the hooks run only after the save commits, a failed request never sends an email, starts a workflow or fires a webhook for data that was rolled back. Loop protection stops workflows from triggering each other forever.

SECTION 6

Module workflows

How each part of the CRM is used, in the order a team usually meets them. Menu paths are shown in violet.

Contacts and companies

Where: **CRM → Contacts**

1. Click **+ New contact**; enter name, email and/or phone, company, owner, lifecycle stage and tags. A possible duplicate (same email or phone) is flagged before saving.
2. Open the contact: the timeline shows everything that happened. Add notes and files, log a call, send email / SMS / WhatsApp, create a deal or task.
3. In the list, filter by tag, owner, stage, source or text; tick several rows to bulk tag, change owner or stage (managers can bulk delete).
4. Companies (CRM → Companies) group contacts and deals and have their own timeline.

CSV import and export

Where: **CRM → Imports (managers)**

1. Upload a CSV, then match each column to a contact field or custom field.
2. Choose what happens to duplicates (skip, update, create) and optionally an owner and tags for everyone imported.
3. Start: the import runs in the background and reports created / updated / skipped rows and the errors per row.
4. Export what the contact list currently shows with **Export**.

Deals and the pipeline

Where: **CRM → Deals**

1. Pick a pipeline; the board shows a column per stage with totals and the weighted forecast.
2. **+ New deal**: title, value, contact / company, owner, expected close date.
3. Drag cards between stages (or onto the won / lost drop zones). Open a deal to mark it Won or Lost with a reason, or Reopen it.
4. Managers set up pipelines, stages, probabilities, rotting days and lost reasons under Workspace → Pipelines.

Tasks, reminders and calendar

Where: **CRM → Tasks / Calendar**

1. Create a task with type (to-do, call, email, meeting, follow-up), priority, due time, reminder, assignee and related record.
2. The reminder email goes out before the due time; the sidebar badge counts tasks due today (red when some are overdue).
3. The calendar shows tasks and deal close dates; drag to reschedule, click a day to add a task.
4. Calendar → Sync to Google Calendar gives each person a private feed link (one-way) for Google, Outlook or Apple Calendar.

Inbox: email, SMS and WhatsApp

Where: **CRM → Inbox**

1. Start a conversation from a contact (Email / SMS / WhatsApp buttons) or reply in the inbox; filter by channel, unread, mine or unknown senders.
2. Templates fill variables such as {{ contact.first_name }}. Email opens and clicks are tracked; WhatsApp read receipts are shown.
3. WhatsApp allows free text only within 24 hours of the contact's last message; after that use a template approved in WhatsApp Manager.
4. **Draft with AI** suggests a reply from the conversation and the workspace's business notes; staff always edit and send it themselves.

Automations

Where: **CRM → Automations**

1. **New workflow**: choose a trigger, optional conditions, then actions in order (see Section 7).
2. Save and switch it on. Use the test run to try it on one record.
3. Each run is logged step by step, so you can see why something did or didn't happen. Automated messages respect opt-outs and quiet hours.

Lead capture and scoring

Where: **Marketing → Web forms / Lead scoring (managers)**

1. Build a form: fields (custom fields too), consent text, owner, stage and tags for new leads; share the link or embed it on your site.
2. Spam is blocked with hidden checks; submissions create or update contacts and can start workflows.
3. Scoring rules: profile rules (points while a condition matches) and behaviour rules (points per event). Grade thresholds turn scores into Hot / Warm / Cold.

Segments, sequences and broadcasts

Where: **Marketing**

1. Segments are saved, always up-to-date lists built from conditions (stage, tags, owner, score, opt-outs, custom fields, inactivity).
2. Sequences: steps with delays (e.g. day 0, 2, 5) by email or SMS; contacts join by hand, from a segment or via a workflow, and leave on reply or unsubscribe.
3. Broadcasts: one email or SMS to a segment, now or scheduled; the result page shows sent, opened, clicked, failed and unsubscribed.

Reports and targets

Where: **CRM → Reports**

1. Filter by period, pipeline and people; KPIs (revenue, win rate, average deal, sales cycle, pipeline, new leads) compare with the previous period.
2. Charts: revenue over time, stage funnel, forecast, lost reasons, lead sources, each with a table view.
3. Leaderboard per rep with CSV export; managers set monthly or quarterly targets; anyone can subscribe to report emails.

API, webhooks and Facebook lead ads

Where: **Workspace → Integrations (admins)**

1. REST API: create a key (shown once, read-only or read/write), read the docs at /api/docs/, connect your website, Zapier/Make or ERP.
2. Webhooks: choose events and a URL; each event is POSTed as signed JSON, retried with backoff, with a delivery log and a test button.
3. Facebook lead ads: add the Meta app secret and pages; subscribe the webhook; leads become contacts and can start the Facebook lead ad trigger.

Help center and AI assistant

Where: **Help & AI assistant · Workspace → AI assistant**

1. Staff ask the Help button how to do something (English or Bangla). Answers come from the help articles and link to them.
2. Admins choose the AI: built-in help only, Ollama (free open-source, on your own server), Groq or OpenRouter (free tiers) or any OpenAI-compatible API.
3. Usage, ratings and errors appear on the settings page; a daily limit per person protects free tiers. Without AI the help center still answers.

SECTION 7

Automation reference

A workflow is **trigger** → **conditions** → **actions**. The trigger decides when a run starts and whether it runs for a contact or a deal; conditions decide whether it continues; actions run top to bottom.

Group	Triggers
Contacts	Contact created (optionally incl. imports) · field changes · lifecycle stage changes · tag added · owner changes · no activity for N days · date field comes round (birthday, renewal)
Deals	Deal created (per pipeline) · moved to a stage · won · lost · expected close date is near · no activity for N days
Activity	Task completed · email opened / clicked / received · SMS received · WhatsApp received · call logged
Leads	Web form submitted · Facebook lead ad submitted (per page) · lead grade changes · new lead not contacted in N hours
Integrations	Incoming webhook (another system POSTs JSON; the contact is found or created)

Group	Actions
Messages	Send email (template or text, as owner or creator) · send SMS · send WhatsApp (approved templates supported) · email a teammate
Tasks	Create a task (type, priority, due in, assignee incl. record owner)
Records	Assign owner (person or round robin) · set lifecycle stage · update a field · add / remove tag · add note · create deal · move deal
Flow	Wait (minutes / hours / days) · continue only if (conditions)
Marketing	Add to / remove from a sequence · change lead score
Integrations	Call a webhook (signed POST to your URL)

Conditions

Conditions can check contact fields (stage, source, owner, tags, score, grade, opt-outs, segment membership, days since created or since last activity, custom fields) and deal fields (title, value, stage, status, owner, tags), with all or any matching.

Example workflows

Goal	Trigger	Actions
Welcome every web lead	Web form submitted	Assign owner (round robin) → send welcome email → create call task due in 1 hour
Never leave a lead waiting	Lead not contacted in 2 hours	Email the manager → assign to another rep → add tag 'Escalated'
Hot lead alert	Lead grade becomes Hot	Notify owner → create task 'Call today' (high priority)
Proposal follow-up	Deal moved to 'Proposal sent'	Wait 2 days → continue only if still in that stage → send follow-up email
Win handover	Deal won	Add tag 'Customer' → create onboarding task → call webhook to accounting
Renewal reminder	Date field 'renewal date', 30 days before	Send WhatsApp template → create task for owner

SECTION 8

A day in the life

How each role typically uses the CRM.

Agent (sales rep)

1. Open the Dashboard: tasks due today, overdue tasks and the pipeline at a glance.
2. Clear the Inbox: reply to email, SMS and WhatsApp (use Draft with AI for speed, then edit).
3. Work Tasks top-down: calls (log the outcome), emails, meetings; tick each one off.
4. Move deals on the board as they progress; mark won or lost with the reason.
5. Add notes after every conversation so the timeline tells the whole story.

Manager

1. Check Reports for the week: revenue vs target, win rate, funnel drop-offs, lost reasons.
2. Use the Leaderboard and filters by team to coach reps; look for rotting deals and overdue tasks.
3. Tune automations, sequences and lead scoring; review run logs when something looks off.
4. Import lists, build segments and send broadcasts; set next month's targets.

Admin

1. Invite members and set roles; organise teams.
2. Keep messaging channels, integrations and the AI assistant set up and tested.
3. Watch System status: database, background jobs and the latest backup.
4. Manage custom fields and workspace settings (time zone, currency).

SECTION 9

First-time setup

The checklist for an admin starting a new workspace. Each line names the screen to use.

#	Do this	Where
1	Create the workspace (sign up) and set time zone and currency	Sign up · Workspace → Settings
2	Invite the team with the right roles; create teams	Workspace → Members / Teams
3	Connect outgoing email (SMTP) and, for replies, an IMAP mailbox	Workspace → Messaging channels
4	Connect SMS (Twilio or a BD gateway) and WhatsApp (Cloud API); send test messages	Workspace → Messaging channels
5	Set sending hours (quiet hours) for automated messages	Messaging channels → Sending rules
6	Adjust pipeline stages, probabilities and lost reasons	Workspace → Pipelines
7	Add custom fields and tags you need	Workspace → Custom fields / Tags
8	Import existing contacts from CSV	CRM → Imports
9	Create message templates (and WhatsApp approved-template names)	Workspace → Message templates
10	Publish a web form; connect Facebook lead ads	Marketing → Web forms · Workspace → Integrations
11	Set lead scoring rules and grade thresholds	Marketing → Lead scoring
12	Turn on key workflows (welcome, speed to lead, deal follow-ups)	CRM → Automations

#	Do this	Where
13	Set monthly targets and subscribe to report emails	CRM → Reports → Targets / Email me reports
14	Choose the AI provider and test it; fill in 'About your business'	Workspace → AI assistant

SECTION 10

Background jobs

Celery beat starts these on a schedule; the worker runs them. Nothing here needs a person.

Job	How often	What it does
Task reminders	every minute	Emails reminders for tasks whose reminder time has come
Inbound email	every 2 minutes	Reads each workspace's IMAP mailbox and threads replies into conversations
Workflow waits	every minute	Resumes workflow runs whose wait has finished (or whose quiet hours ended)
Scheduled triggers	every 15 minutes	Inactive contacts, idle deals, close dates, date fields, speed to lead
Sequences	every minute	Sends the next due sequence step for each enrolled contact
Broadcasts	every minute	Sends scheduled or in-progress broadcasts in batches
Report emails	every 15 minutes	Sends daily / weekly / monthly reports at each person's chosen hour
Webhook retries	every minute	Retries failed outgoing webhooks (1 min, 5 min, 30 min, 2 h, 6 h)
Webhook clean-up	daily	Removes old delivery logs (30 days delivered, 90 days failed)
Facebook lead retries	every 15 minutes	Fetches leads that failed earlier (up to 5 tries)
Backups (production)	nightly	Database dump + uploaded files into the backup volume, kept 14 days

> Messages, imports and webhook deliveries are queued after the request finishes, so pages stay fast. Their outcome (sent, failed with the reason) appears on the timeline.

SECTION 11

Deployment and operations

The same code runs in three ways. Development uses CDN builds and an in-browser Tailwind compiler; the production image compiles CSS and ships every library itself.

Environment	Command	Used for
Local	.venv/Scripts/python manage.py runserver	Quick development with SQLite; background jobs run inline
Docker (dev)	docker compose up -d	Postgres, Redis, a real worker and beat; rebuilds and migrates on every start
Production	docker compose -f docker-compose.prod.yml --env-file .env.prod up -d --build	One server with HTTPS, backups and restart-on-failure

Deploying to a server



Update later: git pull, then the same compose command. Migrations and assets rebuild on their own.

1. Get a Linux server with Docker and point the domain's DNS A record at it.
2. Clone the repository and copy `.env.prod.example` to `.env.prod`; fill in every CHANGE-ME (domain, keys, database password, email server, admin email).
3. Run the production compose command. The image builds (front-end assets compiled), migrations run, then web, worker, beat, Caddy and backups start.
4. Create a superuser if you need the Django admin (at the ADMIN_URL path), then sign up to create the first workspace.
5. To update later: `git pull` and the same command again.

Backups and restore

- Every night at BACKUP_HOUR the backup service writes a database dump and a tarball of uploaded files; 14 days are kept. System status turns red if the newest backup is over 36 hours old.
- Copy the backups off the server as well (another machine or cloud storage). Keep FIELD_ENCRYPTION_KEY with them: without it, saved email, SMS, WhatsApp and AI keys can't be read.
- Restore: stop web, worker and beat; run the restore command with the dump's name and --yes; start the stack again.

```
# shorthand below:  dc = docker compose -f docker-compose.prod.yml --env-file .env.prod
dc exec backup sh /backup.sh once                                  # back up now
dc exec backup sh /backup.sh list                                  # list backups
dc stop web worker beat                                            # before a restore
dc exec backup sh /backup.sh restore crm-YYYYmmdd-HHMMSS.dump --yes   # replace the database
dc up -d                                                           # start again
```

Security in production

- HTTPS only through Caddy (automatic certificates), HSTS, secure cookies, clickjacking and content-type protection; Django's deploy check passes.
- The app refuses to start without a strong secret key, the domain in ALLOWED_HOSTS and the field encryption key.
- Login throttling: 5 failed logins for one email from one IP lock it for 15 minutes; password-reset emails are limited per IP.

- Secrets (SMTP / SMS / WhatsApp / Facebook / AI keys) are encrypted in the database; API keys are stored only as hashes and shown once.
- Outgoing webhooks refuse private and localhost addresses; Postgres and Redis are not reachable from outside the server.
- Strict workspace isolation is covered by tests for every module; CI runs tests, the migrations check, the deploy check and the image build on every push.

SECTION 12

Demo access and project history

Demo access

The **Demo Traders** workspace comes with sample companies, contacts, deals (won, lost and open), tasks, a team and a monthly target. Create or refresh it with `python manage.py seed_demo` (in Docker: `docker compose exec web python manage.py seed_demo`).

Role	Login email	Password
Admin	admin@demo.localhost (the DEMO_EMAIL value)	DEMO_PASSWORD in the project's .env file
Manager	manager@demo.localhost	same as above
Agent	agent@demo.localhost	same as above

Passwords are deliberately not printed in this document. Read them from .env, or change DEMO_PASSWORD and run `seed_demo --reset-password`. Local: http://localhost:8020 · Docker: http://localhost:8000.

Project history

Phase	Delivered	Commit
0	Foundation: email login, workspaces, roles, teams, Celery, Docker	80bdad7
1	Contacts, companies, tags, custom fields, notes, files, timeline, CSV import / export	dc16d54
2 + 3	Pipelines, deals, kanban, tasks and reminders · email, SMS, inbox, tracking, calls, Select2 / DataTables / FullCalendar	68c1f09
4	Automation engine: triggers, conditions, actions, waits, builder, run logs, webhooks	46ab1b8
5	Lead and communication automation: forms, scoring, segments, sequences, broadcasts, consent, quiet hours	e0ebd65
6	Sales and reporting: dashboard, funnel, forecast, leaderboard, targets, report emails, CSV	7483633
7	REST API, webhooks, WhatsApp, Facebook lead ads, calendar feed, sidebar groups, AI help assistant	63dc413
8	Hardening and deploy: production stack, HTTPS, compiled assets, login throttling, backups, CI, demo seed	762ca57

Next steps outside the code: deploy to the production server, confirm the first CI run on GitHub, set up off-server backup copies, and connect the live WhatsApp, Facebook and AI accounts.